

AUTONOMOUS MULTIMODAL VIDEO & AGENT MEMORY SPECIFICATION

ZERO-COST INSTAGRAM REELS AUTOMATION · WORD-BY-WORD SUBTITLES · ZERO-TOKEN REPO MEMORY

1. Executive Architecture Overview

System Summary: This master document specifies a production-grade, **\$0.00 zero-cost multimodal pipeline** for autonomous Instagram Reels generation combined with a **Zero-Token Persistent Context & Memory System** for AI pair-programming agents (Antigravity IDE, OpenCode, Claude Code, Cursor). The content pipeline automates viral scripts, studio-grade neural voiceovers, FLUX.1 9:16 visuals, word-by-word kinetic highlight subtitles, and FFmpeg video compositing. The memory system stops agents from burning 40k–120k tokens on restarts by utilizing compressed AST symbol maps and active state ledgers.

2. Zero-Cost Multimodal API & Tool Matrix

Modality	Provider / Library	Free Access Endpoint	Quotas & Rate Limits	Card?
Text & Scripts	OpenRouter (:free tier)	openrouter.ai/keys (glm-5.3-flash, gemini-2.0-flash)	Uncapped lifetime requests on free models. RPM rate limits.	No
Voiceover (TTS)	Microsoft Edge-TTS	pip install edge-tts (en-US-ChristopherNeural)	100% Free, zero keys, zero quotas, studio-quality neural audio.	No
Word Timestamps	Faster-Whisper (int8)	Local CPU/CUDA inference	Word-level millisecond timestamps without external API dependency.	No
9:16 Visuals	Pollinations.ai / FLUX.1	image.pollinations.ai/prompt/...	Zero authentication required, direct 1080x1920 FLUX.1 generation.	No
Video Compositing	FFmpeg + libass	Local CLI binary	Hardware-accelerated scale, Ken Burns zoompan, burned subtitles.	No
Social Publishing	Meta Graph API v21+	developers.facebook.com	Official Reel container publishing, up to 50 posts per 24h per account.	No

3. Automated Instagram Reels Pipeline & Kinetic Subtitles

The Reels engine executes a 5-step automated workflow: **1. Script Gen** (OpenRouter extracts high-retention hook & under-45-word narration); **2. Audio Synth** (Edge-TTS generates 44.1kHz MP3 with natural speech tempo); **3. 9:16 Visual Retrieval** (Pollinations FLUX.1 fetches vertical asset); **4. Kinetic Subtitle Compilation** (Faster-Whisper extracts word timestamps and writes ASS karaoke tags with bright yellow highlight &H0000FFFF; and font scale pulse \fscx118\fscy118); **5. FFmpeg Compositing** (Executes smooth Ken Burns zoompan and libass subtitle burn into progressive H.264 MP4).

Core Pipeline Execution Logic (pipeline_reels.py):

```
# 1. Script -> OpenRouter glm-5.3-flash:free
completion = client.chat.completions.create(model='z-ai/glm-5.3-flash:free', ...)
# 2. Voiceover -> Microsoft Edge-TTS (Zero keys, uncapped)
communicate = edge_tts.Communicate(text, voice='en-US-ChristopherNeural', rate='+6%')
# 3. Subtitles -> Faster-Whisper word timestamps to ASS karaoke formatting
line = f'\c&H0000FFFF;\fscx118\fscy118}{active_word}{\fscx100\fscy100\c&H00FFFFFF;}'
# 4. FFmpeg Video Compositing with Ken Burns zoompan filter
ffmpeg -loop 1 -i bg.jpg -i audio.mp3 -filter_complex 'zoompan=z=min(zoom+0.0018,1.2):d=900:s=1080x1920,subtitles=subs.ass'
out.mp4
```

4. Zero-Token Persistent Memory Architecture for AI Agents

The Problem: When restarting Antigravity, OpenCode, Claude Code, or Cursor, the LLM context window resets. Naive agents immediately run recursive directory scans (find ., ls -R) and parse whole source files, consuming **40,000 to 120,000 tokens** before doing any useful work.

The Solution: A 4-Tier Memory & Context Retention Architecture that keeps initial resume overhead **under 3,000 tokens**:

Tier	Component & Artifact	Function & Token Impact
Tier 1	AST Architecture Map .agents/memory/ARCHITECTURE_MAP.md	Extracted class and function signatures only. Gives the agent complete structural visibility across the repo without loading file bodies. Total size < 3-5 KB (~1,000 tokens).
Tier 2	Active Session Ledger .agents/memory/ACTIVE_SESSION.md	Stores current project stage, completed tasks, active files, and immediate next action. Prevents repetitive questioning and ensures continuous context alignment across restarts.
Tier 3	Local MCP Memory Server codebase-memory-mcp / SQLite	Targeted semantic search and call-graph tracing (trace_path, search_graph). Replaces brute-force recursive file grep with graph queries.
Tier 4	Prompt Caching Strategy OpenCode setCacheKey + Prefix	Maintains static system prefixes and rules to achieve 90%–98% prompt caching hit rates on Gemini, Claude, and OpenRouter, minimizing latency and billing.

5. Agent Startup Protocol (`/start` Workflow)

Deploy .agents/workflows/start.md in your repository. Upon launching Antigravity or OpenCode, typing **/start** triggers:

- Read State:** Agent loads .agents/memory/ACTIVE_SESSION.md only.
- Structural Consultation:** Agent checks .agents/memory/ARCHITECTURE_MAP.md for symbol locations.
- Strict Rule Enforcement:** No directory trees, no recursive scans, no full-file bulk loading.
- Checkpoint Protocol (/checkpoint):** Before shutting down, run python scripts/generate_repo_map.py and update ACTIVE_SESSION.md with completed milestones.

6. Implementation Checklist & Tech Stack Roadmap

Phase	Milestone Description	Status / Verification
Phase 1	Setup Free Multimodal Engine (OpenRouter, Edge-TTS, Faster-Whisper, FFmpeg)	Ready in pipeline_reels.py
Phase 2	Kinetic ASS Subtitle Generator with 3-word chunks and yellow highlighting	Verified with libass styling
Phase 3	Meta Graph API 2-step media container publishing for Instagram Reels	Ready in publish_instagram.py
Phase 4	AST Codebase Compressor (scripts/generate_repo_map.py)	Tested: Generated 20KB index
Phase 5	Active Session Ledger & /start Slash Command Workflow	Deployed in .agents/